



MA 313: Formal Languages and Automata Theory

Course Code: MA 313

Course Name: Formal Languages and Automata Theory

L-T-P-C: 3-0-0-3

Students intended for: B.Tech Maths and Computing

Elective or Core: Core

Prerequisites: CS 202

Mutual Exclusion: CS 304

Approval: 59th BoA

Course Contents

- **Regular languages:** DFA, NFA, Subset construction, Regular, Pumping Lemma, DFA state minimization, Myhill-Nerode relations and theorem. [10 Lectures]
- **Grammars:** Production systems, Right linear grammar and Finite state automata, Context free grammars, Normal forms, Pumping Lemma for CFLs, Subfamilies of CFL, Derivation trees and ambiguity. [6 Lectures]
- **Pushdown automata:** Acceptance by final state and empty stack, Equivalence between pushdown automata and context-free grammars, Closure properties of CFL, Deterministic push-down automata, the CKY algorithm. [10 Lectures]
- **Turing machines:** Techniques for Turing machine construction, Generalized and restricted versions equivalent to the basic model, Universal Turing machine, Recursively enumerable sets and recursive sets. [9 Lectures]
- **Decidability:** Decidable and undecidable problems, Reduction, Post's correspondence problem, Rice's theorem, decidability of membership, emptiness and equivalence problems of languages. [7 Lectures]

Text books:

1. J. E. Hopcroft, R. Motwani and J.D. Ullman, **Introduction to automata theory, languages and computation**, 3rd Edition, Pearson, 2006.
2. M. Sipser, **Introduction to the Theory of Computation**, 3rd Edition, Cengage Learning, 2012.

References:

1. D. C. Kozen, **Automata and Computability**, Springer, 1997.
2. E. A. Rich, **Automata, Computability and Complexity: Theory and Applications**, Pearson, 2007.
3. Peter Linz, **An introduction to formal language and automata**, 3rd Edition, Narosa publishing house, 2002.